Wind load Analysis For Rolling Shutter
By as per Indian Standards IS 875 Part III 1987

Project : Godrej & Boyce
Client : RBCC Ltd.
Date : 06/10/2016

By

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Wind Load Analysis as per IS 875 - III

- Design Wind Speed $Vz = V_b * k_1 * k_2 * k_3$

  where V_b = Basic Wind Speed in m / s.

  k_1 = Risk Coefficient (clause 5.4.1) = 1

  k_2 = THS Factor (clause 5.3.2) = 1

  k_3 = Topography Factor (clause 5.3.3) = 1

  V_b = Maximum = 47 m/s

- Wind Speed Vz= 47 m/s

- Terrain Category: 2 (Open terrain with well-scattered obstructions having heights between 1.5m to 10m)

- Class of Building: A(Structures having max. dimensions is less then 20m)

- k1, k2 & k3 constants are derived from IS-875, based on location height of structure & geographical factor. These factors are required for tall structures, where we need to consider varying pressure condition.

Wind Load Analysis as per IS 875 - III

- Design wind pressure = $0.6 \times V_z^2$

Where Pz = Design wind pressure

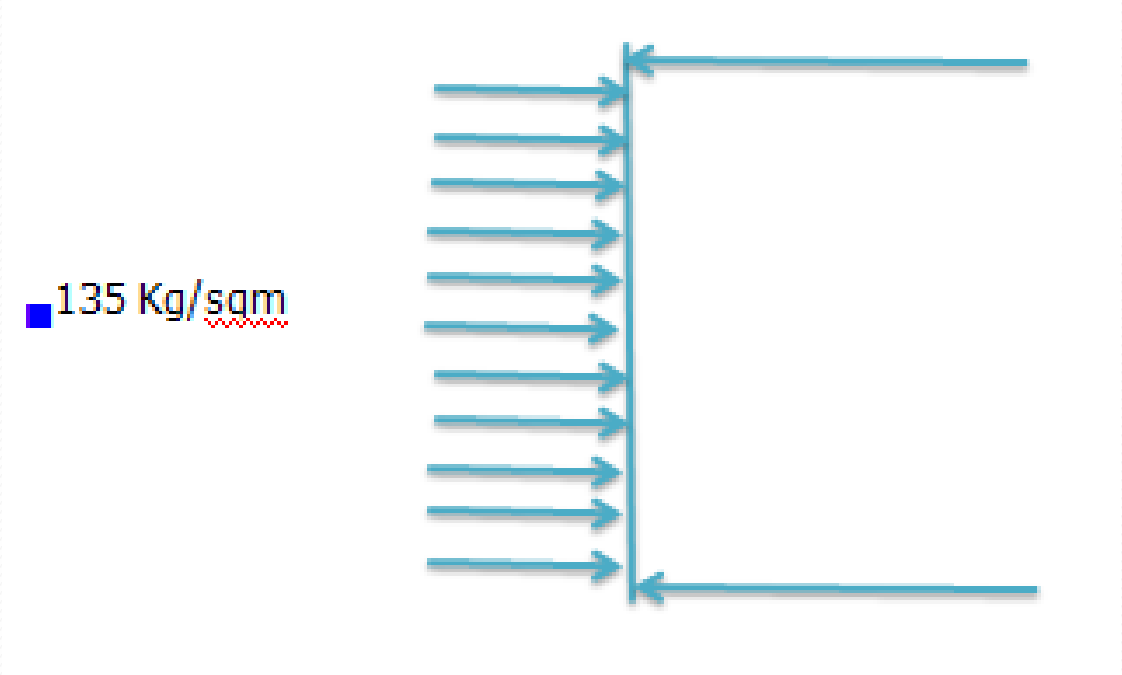
Vz = Design wind velocity

Wind Pressure = $0.6 \times (47)^2$

Wind Pressure = 1325.4 N/m^2

Wind Load Analysis as per IS 875 - III

- **Case I - Simply supported beam with uniform pressure.**



Contents:

A) Bending Moment

B) Deflection

C) Maximum Stress

D) Side guides for curtain

A)Bending Moment

Door is supported on two opposite sides

Max B.M

= 0.056*Span*Span*Design Pressure

B)Deflection

Max. deflection will be at the centre

$$D_{max} = (5x M x We^2) / (384 x E x I)$$

Where , M = Bending Moment of 75mm Curved type Slat

E = Modules of Elasticity for CRS = 210GPa (KN/mm2)

I = Moment of Inertia of slat ($b \times d^3 / 12$)

We = Effective Width of Shutter

w = clear width of shutter

C)Max Stress

Max. Stress= We^2xpressure on stripx1000xC/12x I
Where C= centre of gravity I= moment of inertia , We= effective width

Permissible Stress = 241/FOS
= 241/1.5
= 160.66 N/sq.mm

D)Side Guides for Curtain panel

Side Guides required = w / 60

Where , w = Width of shutter

Design calculations for small shutter of Size: 3000mm(w) x 2800mm(H)

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1. Wind Analysis for rolling shutter of size 3000(w) x 2800(h)mm

- Design Wind Speed $Vz = V_b * k_1 * k_2 * k_3$

  where V_b = Basic Wind Speed in m / s.

  k_1 = Risk Coefficient (clause 5.4.1) = 1

  k_2 = THS Factor (clause 5.3.2) = 1

  k_3 = Topography Factor (clause 5.3.3) = 1

  V_b = Maximum = 47 m/s

  wind Speed Vz= 47 m/s

Design wind pressure = 0.6x 47x47= 1325.4 N/m2

Bending force F = 0.056 x span x span x Design pressure

= 0.056 x3.1x3.1x1325.4

= 713.27 N

Bending Moment M= Fx0.075m (M of 75mm slat)

=713.27x0.075

=53.49 Nm

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- Assuming thickness of slat 1.2mm & minimum width as 15mm & depth as 75mm.

I=(15x15x15x75)/12 – (12.6x12.6x12.6x74.6)/12
=8658.08 mm4

E= modulus of elasticity for CRS= 210 GPA
= 2.1 x10^5 N/mm2.

Max. deflection= 5x53.49x1000x(3.1x1000)^2/(384x2.1x10^5x8658.08)
= 3.68mm

Permissible deflection = 3100/300
=10.33mm

Deflection is within limit.

Max. stress = 3.15x3.15x0.075x1325.4x10000x7.5/(8658.07)
= 71.19 N/mm2.

Permissible stress = (241 N/mm2) /1.5
=160.66 N/mm2

Hence, design is Safe.

Guide side required = 3100/60 =51.66

Hence we are using 100mm guide.

Design calculations for large shutter of Size: 8000mm(w) x 8000mm(H)

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2. Wind Analysis for rolling shutter of size 8000mm(W) x 8000mm(H)

- Design Wind Speed $Vz = V_b * k_1 * k_2 * k_3$

  where V_b = Basic Wind Speed in m / s.

  k_1 = Risk Coefficient (clause 5.4.1) = 1

  k_2 = THS Factor (clause 5.3.2) = 1

  k_3 = Topography Factor (clause 5.3.3) = 1

  V_b = Maximum = 47 m/s

  wind Speed Vz= 47 m/s

Design wind pressure = 0.6x 47x47= 1325.4 N/m2

Bending force F = 0.056 x span x span x Design pessure

= 0.056 x8.2x8.2x1325.4

= 4990.71 N

Bending Moment M= Fx0.075m (M of 75mm slat)

=4990.71x0.075

=374.30Nm

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- Assuming thickness of slat 1.2mm & minimum width as 15mm & depth as 75mm.

I=(15x15x15x75)/12 – (12.6x12.6x12.6x74.6)/12
=8658.08 mm4

E= modulus of elasticity for CRS= 210 GPA
= 2.1 x10^5 N/mm2.

Max. deflection= 5x374.3x1000x(8.2x1000)^2/(384x2.1x10^5x8658.08)
= 180.23mm

Permissible deflection = 8200/300
=27.33~ 28mm

Max. stress = 8.2x8.2x0.075x1325.4x1000x7.5/12x(8658.07)
= 482.49 N/mm2.

Permissible stress = (241 N/mm2) /1.5
=160.66 N/mm2

Guide size required= 8000/60
= 133.33mm

Hence we are using 150mm guides.

Wind Load Analysis
for different wind speed

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Wind Load Analysis for different wind speed

Size of shutter W x H in mm	Wind speed in Km/ph	Allowable Deflection in mm	Deflection in mm
3000 x2800	30 (8.33 m/s)	10	0.11
	60 (16.66 m/s)	10	0.45
	120 (30.33 m/s)	10	1.53
	170 (47 m/s)	10	3.68

Size of shutter W x H in mm	Wind speed in Km/ph	Allowable Deflection in mm	Deflection in mm	Deflection With grill In mm
8000x8000	30 (8.33 m/s)	28	5.66	3.39
	60 (16.66 m/s)	28	22.66	13.59
	120 (30.33 m/s)	28	75.05	45.03
	170 (47 m/s)	28	180.23	108.01

Note-By placing grill of size 8mx2.2m will reduce deflection by approximately 30 to 40%.

Thank You

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